UNIT 2: SHAPES AND REGIONS

SESSION 5: ACTIVE CONTOURS

SLO 1: EXPLAIN THE CONCEPT AND WORKING PRINCIPLE OF ACTIVE CONTOURS FOR OBJECT BOUNDARY DETECTION IN IMAGES.

Process Arrangement

Arrange the following steps in the correct order for active contour operation.

- The contour gradually moves toward the object boundary.
- Initialize a contour near the object.
- Energy is minimized using internal and external forces.
- The contour stabilizes at the object boundary.

Correct Order:

1. Initialize a contour near the object.
2. Energy is minimized using internal and external forces.
3. The contour gradually moves toward the object boundary.
4. The contour stabilizes at the object boundary.

Activity 4 – Concept Mapping

Match the following:

Term	Description
Active Contour	A. Initial curve placed near the object
Snake Model	B. Boundary detection technique
External Force	C. Pulls contour toward image edges
Internal Force	D. Maintains smoothness of contour

Activity 5 – Think and Apply

Imagine detecting the boundary of a cell in a medical image.

1. Why is active contour useful for this task?

Active contour is useful for detecting the boundary of a cell because it can gradually move toward cell boundary using internal and external forces. It helps accurately identify and trace the shape in image.

2. What might happen if the initial contour is placed far from the object?

If the initial contour is placed far from the object, it may not reach the correct boundary. The contour can get stuck at an incorrect location or may require more iterations to converge to the actual object boundry.